

Understanding network equipment reliability

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This paper reviews the concepts and techniques that have been adopted by BT in order to understand network equipment reliability. The ideas presented are largely generic, in the sense that they can be applied irrespective of whether the equipment is intended for a core network or an access network application. In particular, the paper examines the relationship between out-turn reliability and the reliability that might be guaranteed by an equipment manufacturer. The relevance of confidence limits to reliability modelling and in-service reliability monitoring is also discussed, and ideas presented on how reliability monitoring can be applied to resilient networks. It is shown that in competitive situations, there are good reasons why reliability modellers should use failure rate data whose upper confidence limit is 50%. In contrast, it is shown that for practical reasons, in-service monitoring schemes should not be based on specific confidence levels.

1. Scope

The increasing penetration of active electronics in the access network means that service reliability is becoming less dominated by external plant contributions. It is therefore important that network operators have credible practices for understanding and influencing access network equipment reliability. This paper reviews BT's practices in this area, many of which were developed for core network applications but are now increasingly being applied to access electronics. The material presented is complementary to many previous access network reliability studies, which have tended to concentrate on plant components, e.g. Mellis [1].

After presenting some introductory reliability concepts, this paper examines the relationship between out-turn and guaranteed reliability of field replaceable units (FRUs), and shows how the confidence level of FRU failure rates affects the results of reliability modelling. The paper then discusses some related aspects of in-service reliability monitoring schemes. Specifically, the following questions are addressed.

- By what factor does out-turn reliability exceed guaranteed reliability?
- What confidence levels should be used for FRU failure rates when modelling network reliability?
- What confidence levels should be used for FRU failure rates when establishing reliability demonstration plans or in-service monitoring (ISM) schemes?
- Should ISM schemes be based on FRU reliability or on service reliability, and how can they best be applied to resilient networks?

Data to answer the first two questions has been incorporated into the BT reliability databases that are used to support network reliability modelling and feasibility studies on network design options. This will help optimise future network designs, particularly those involving resilience. Answers to the second two questions have been incorporated into contracts for wavelength division multiplexing (WDM) equipment, recently deployed on pilot routes in the UK core network.

2. Reliability concepts

Several failure rate parameters may be used to describe the reliability of an FRU, and as they generally take different numerical values it is important to understand the differences between them. These parameters are discussed in the context of equipment design and procurement activities.

As part of the design process, manufacturers normally make a standard assessment of the reliability of each FRU type. To do this, they maintain a database of component failure rates, where failure rates of individual components are listed at a known upper confidence level (60% is commonly used, but other confidence levels are not precluded). The failure rates for the components on an FRU are then added either by parts count analysis or by parts stress count analysis [2] to give the overall FRU failure rate. Provided the confidence level of the component failure rates is greater than 50%, it turns out that the confidence level of the FRU failure rate is higher than the confidence level of the component failure rates [3]. However, reliability predictions from component databases are often pessimistic, and cannot be relied upon to give an accurate indication of out-turn reliability.

As part of the tendering process, BT normally states a service reliability target and asks manufacturers to guarantee¹ a minimum FRU reliability, below which they undertake to investigate and correct any deficiencies. The failure rate parameters for an FRU design are therefore:

- design target (expected value): the long-term steady-state reliability the manufacturer expects the design to realise — this figure should be regarded as an absolute value (i.e. not subject to any confidence limits), and is not usually declared to BT,
- guarantee: the minimum reliability (maximum failure rate) the manufacturer is prepared to guarantee to customers — this should also be regarded as an absolute value,
- prediction: the standard assessment of reliability according to whatever component level database the manufacturer operates — this is normally associated with a specific confidence level, and may or may not be equal to either the design target or the true value,
- true value: the actual reliability realised in service, based on failure of the entire population — this is an absolute value, not subject to any confidence limits, and, in practice, is generally never known because only a minority of units would be expected to fail within the service lifetime.

Manufacturers usually build in a 'safety margin' between their design target and the guarantee, typically allowing a factor of about 2 (i.e. guaranteed failure rate = 2 × expectation). BT then normally operates some form of reliability monitoring to compare out-turn reliability with the guaranteed figure. Typical relationships between the various parameters might be as shown in Fig 1. This paper assumes that FRU failure rates are constant with time.

¹ In this paper, the term 'guaranteed' reliability means either the minimum reliability that is contractually guaranteed by the equipment manufacturer, or, failing such a guarantee, the term is taken to mean the target reliability agreed between the customer and the manufacturer.

2.1 Confidence levels

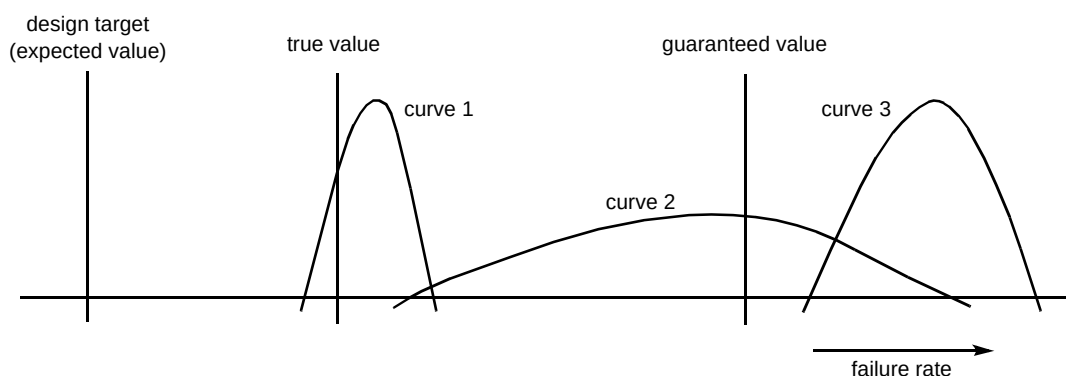
Whenever there is imprecise information available about the equipment, confidence levels should always be quoted to indicate the degree of uncertainty in the reliability parameter. For example, confidence levels should be quoted for a reliability prediction, or for in-service reliability measurements based on the failure of only a proportion of the population. In this latter case, although there is precise information on the failures to date, it is normally the reliability of the whole population that is of interest, and the reliability of the population that has not yet failed can only be estimated. The confidence in the reliability estimate of the whole population will improve as the number of failures increases.

In some circumstances the concept of confidence limits does not apply, and a precise value can be assigned to a reliability parameter. In these circumstances an absolute number can be quoted, which cannot be 'varied' by some debate about confidence levels. For example, customers may specify the maximum failure rate they are prepared to tolerate, or manufacturers may guarantee the maximum failure rate they expect the product to exhibit. Similarly, if the entire population of equipment had failed, it would be possible to make a precise statement about its reliability.

3. Relationship between out-turn and guaranteed reliability

This relationship is of interest because it enables a more realistic assessment of likely network reliability than would be obtained by using guaranteed or target reliability figures which tend to be conservative. Despite the relationship being a complex function of many variables, a pragmatic analysis can be easily undertaken.

In-service reliability measurements were obtained from four of BT's suppliers, for a total of six different types of equipment. The data set covered 576 different types of FRU, and was chosen so that failure rates could be calculated according to a simple formula and also calculated



Curves 1 and 2 are estimates of the true failure rate based on failure of a proportion of the equipment population. The number of failures for curve 1 is greater than the number of failures for curve 2. Curve 3 is an estimate of the true failure based on a reliability prediction.

Fig 1 Typical relationship between FRU reliability parameters.

at 50% confidence². Figure 2 shows the number of FRU types having particular values of the ratio 'guaranteed failure rate/50% confidence failure rate'. The graph shows a believable curve up to a ratio of about 10, by which time around 90% of the FRU types have been captured. The remaining 10% of FRU types show ratios extending out to beyond 100. In some cases these large values will be due to inaccuracies in the base data (e.g. under-estimating the number of failures, or over-estimating the time in service). Neglecting such FRUs from the analysis, the average value of the 'guaranteed/50% confidence' failure rate ratio is 2.17. A factor of 2.0 has therefore been incorporated into the reliability databases³ to reflect the improvement in out-turn reliability over guaranteed reliability.

4. Reliability modelling

Reliability modelling provides an input to network design studies by highlighting those features having most impact on the overall reliability budget, and by influencing design activity to investigate the most rewarding scenarios. Inputs to the models are equipment and infrastructure failure rates and their associated restore times. Outputs from the models are service failure rates and downtime (unavailability), sometimes expressed as a 'probability of availability', rather than as a single long-term average [4]. Such modelling is also used to indicate whether a network design is likely to meet its quality of service (QoS) requirement. Until recently, the BT network reliability database used values for FRU failure rates that were exclusively based either on the simple in-service measures or on the expected reliability.

Network designers and marketing personnel tend to accept reliability modelling results without questioning their statistical basis, and therefore the reliability modeller should use the most appropriate confidence level for each exercise. The reliability model most commonly used within BT's network transport area [5] produces outputs of the 'average reliability expectation' for particular customer

groupings. In practice, some customers will experience better reliability, and some will experience worse reliability.

This 'average reliability expectation' is consistent with typical QoS targets. For example, a service availability target of 0.999 does not mean that we expect all customers to always experience at least this availability (because there is a small but finite probability that customers will actually receive a very low availability), but that we expect customers to experience at least this availability **on average**.

In a competitive environment, if BT adopted a cautious marketing stance and published conservative reliability targets, it would lose business to the competition. Equally, BT should not publish unduly optimistic reliability targets that there is little chance of meeting.

The above arguments suggest that in most cases⁴ BT should use confidence levels that are consistent with this average approach, and intuitively a choice of 50% confidence seems reasonable. In fact, there are two supporting arguments as to why this is a good choice.

- Most modelling uses mean values of parameters such as the mean time to repair (MTTR). In particular, the MTTR parameter often has a relatively high value for infrastructure faults, and these make a significant contribution to overall unreliability. It is therefore reasonable to use FRU confidence levels that are consistent with this 'mean' approach.

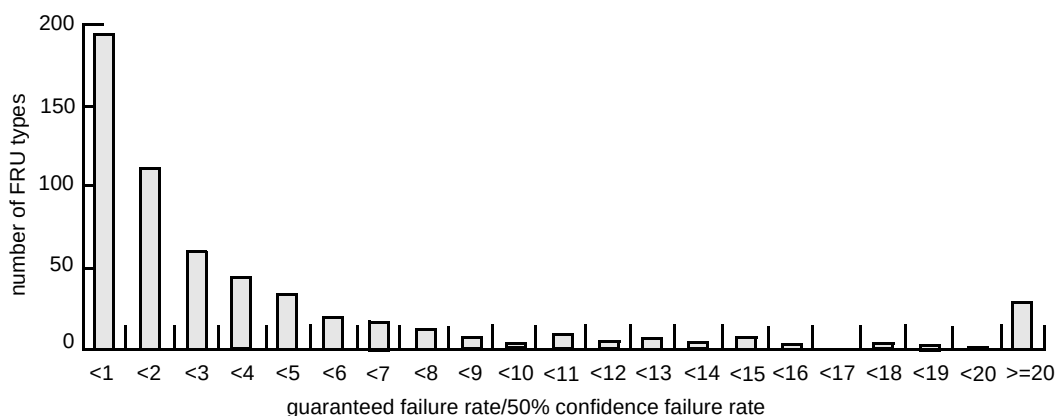


Fig 2 The number of FRU types having particular values of the ratio 'guaranteed failure rate/50% confidence failure rate'.

² The simple formula used is mean time between failures (MTBF) = cumulative time in service (years)/number of failures, or failure rate (FITS) = $1E9 \times$ number of failures/cumulative time in service (hours).

³ The reliability database most commonly used within the BT Network Transport Engineering Centre is available within BT on the BT intranet at <http://networks.intra.bt.com/ntsi/areas/reliab/index.htm>

⁴ In some situations (e.g. safety critical systems such as air traffic control networks) a more conservative approach could well be appropriate.

- The confidence level of a service path failure rate remains at 50% for series of different FRUs, provided that the FRU failure rates are also at 50% confidence. Where FRU failure rates are at confidence levels other than 50%, the confidence level in the service path failure rate departs even more from 50%, and this has implications for both network design and marketing.

Figure 3 shows the variation of the confidence level in the service path failure rate against the confidence level in the FRU failure rate for a service path comprising a series of four different FRU types. If each FRU type has a failure rate of 4000 FITs at 60% confidence, the confidence level associated with the 16000 FITs service path failure rate is increased to around 70%. Conversely, if the FRU failure rates were at 40% confidence, the confidence level in the path failure rate is reduced to around 30%. FRU confidence levels >60% or <40% produce an even more marked effect on the confidence level in the service path failure rate. Only if the FRU failure rates are at 50% confidence, does the path failure rate remain at 50% confidence. The slope of the lines in Fig 3 is due to the finite width (standard deviation) of the FRU failure rate estimate distribution. A similar graph can be produced to show how the number of series FRUs in the service path affects the confidence level in the service path failure rate. This behaviour does not apply to series of the same FRU type because, in this case, the FRUs are no longer independent of one other.

5. In-service monitoring schemes

BT often conducts an acceptance test or reliability demonstration test prior to roll-out of large quantities of equipment, and/or conducts routine reliability monitoring once the equipment is in service. In the first case, BT is concerned with making a single decision to accept/reject a population of equipment based on the results of the test. In the second case, BT is interested in a long-term comparison between measured and guaranteed reliability, where failure to meet the guarantee requires remedial action by the manufacturer and may be linked to contractual penalties. It is important not to confuse these two different situations.

Reliability demonstration tests can either be constructed from first principles or a standard plan can be selected [6]. Plans fall into one of two basic categories — single decision point plans or sequential demonstration tests. The sequential demonstration test is the most common because it achieves a result in a shorter time. The basic premise of the plans is to demonstrate that the reliability of the equipment is **better** than a certain limit, rather than to determine its actual reliability. The plans require the manufacturer to accept a small risk that the equipment will be rejected (even though it is actually reliable enough) and the consumer to accept a small risk that the equipment will be accepted (even though it is actually not reliable enough). These risks arise through making projections about the true failure rate of the equipment on the basis of limited information as a result of only allowing a proportion of the equipment to fail. Given that the risks and levels of acceptable/unacceptable reliability can be agreed, it is possible to generate a plan for the number of failures against time. The question of confidence limits does not formally arise in the construction of the plan. In the author's experience, formal sequential tests tend to be reserved for equipment such as narrowband switch line cards, payphone products, etc.

In-service reliability monitoring schemes have been operated by BT for a number of years, based on fairly pragmatic criteria. The schemes require the manufacturer to calculate equipment reliability from a knowledge of the number of returns of each FRU type, and of their in-service duration. Results are frequently expressed over specific time windows, e.g. last 3 months, last 12 months, and cumulative. Because the measurements are based on failure of only a proportion of the equipment population, confidence limits should strictly be associated with the results. However, as far as the author is aware, all such schemes are based on simple formulae for determining FRU reliability. These calculations give results whose confidence level varies according to the number of failures, and do not give meaningful results in the case of zero failures. Figure 4 shows that these simple calculation methods produce confidence levels that approach 50% as the number of failures increases, but that early estimates of whole

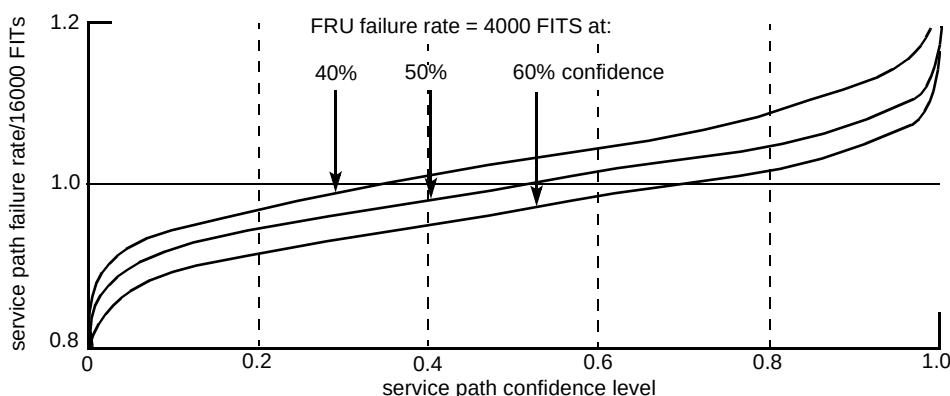


Fig 3 Effect of FRU confidence limit variation on service path confidence limit.

population failure rate are optimistic, even for reasonable numbers of failures. BT operates ISM schemes of this type for equipment such as core transmission systems, System X, TPN and DACS. The concept of supplier/consumer risk does not apply to monitoring schemes of this type.

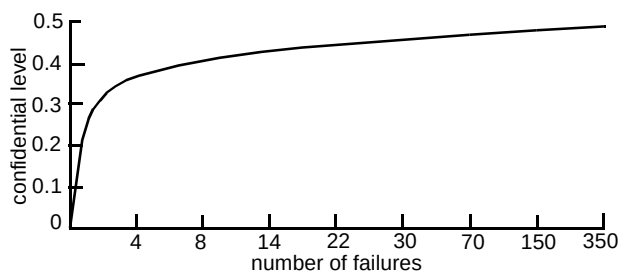


Fig 4 Variation in confidence level with number of failures based on simple measures.

5.1 Confidence levels

It is useful to consider whether ISM schemes should be based on the 'simple measures' or whether a fixed confidence level scheme would be more appropriate. However, manufacturers would recognise that fixed confidence level schemes were more pessimistic (Fig 5), and would increase their FRU failure rate guarantees so as not to compromise their chances of meeting the guarantee. The net result would be that BT would have no more control over manufacturers than it does now, particularly since 'special rules' often apply to the early periods of ISM schemes to protect manufacturers from rogue failures. In addition, some manufacturers may be reluctant to implement a fixed confidence level ISM scheme if changes were required to their fault reporting databases. On balance therefore, it is probably sensible for BT to retain the existing simple calculation methods, despite their extreme optimism during the early stages of the plans.

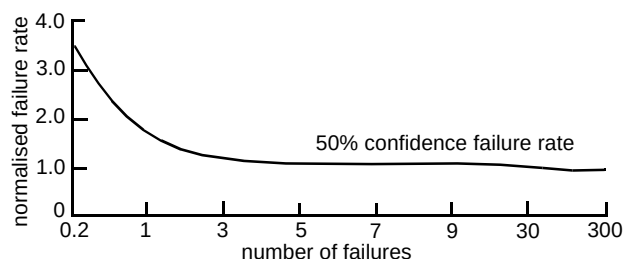


Fig 5 Variation in ratio of 50% failure rate to true failure rate with number of failures.

5.2 Non-resilient networks

There are two ways in which an ISM scheme can be established; these are illustrated by considering the five FRU types arranged in a non-resilient service path in Table 1.

If a manufacturer guaranteed the maximum failure rate of each FRU type, the maximum path failure rate would be 8000 FITS. However, it could be argued that because the manufacturer would expect to beat each FRU guarantee, it

would be very unlikely that the path failure rate would ever exceed 8000 FITS. Therefore it may be possible to negotiate a lower path failure rate guarantee in return for a less detailed ISM scheme, e.g. a maximum 7000 FIT path failure rate limit might be agreed with the manufacturer in return for setting up the ISM scheme to monitor the path failure rate rather than FRU failure rates.

Table 1 Difference between FRU guarantees and service path guarantees.

Item	FRU guarantee	Path guarantee	FRU out-turn (say)
FRU 1	2000		1000
FRU 2	1500		900
FRU 3	2000		1000
FRU 4	1000		1200
FRU 5	1500		900
Path	8000 total	7000 total	5000 total

This raises the question of whether the tighter limit would make any real difference to the out-turn reliability. Given that manufacturers should not normally fail an ISM scheme, it is debatable which approach would give the better reliability. However, experience has shown that there are practical advantages in being able to control manufacturers at the FRU level. Given the above arguments, BT should continue to operate FRU-based ISM schemes.

5.3 Resilient networks

The above argument is adequate for equipment that has limited (or no) internal redundancy and which does not provide service over resilient network paths. However, where this is not the case, although an FRU-based ISM scheme gives some control over maximum FRU failure rates, it will provide very limited control over service failure rates. Even if a service reliability model⁵ had been agreed with the manufacturer, and it was argued (just on the basis of the model) that the guarantee was not being met, the manufacturer would argue that the model was pessimistic rather than try to improve their equipment. Clear evidence would need to be produced to show that the guarantee was not being achieved in order to get any investigative action from the manufacturer. In such cases, although an FRU ISM scheme is a sensible minimum, it needs to be supported by measurements of actual service reliability, which can then be compared against an agreed limit. Measurement of traffic affecting faults requires an operational BT process, centred on the network operation's units. Such a scheme has been established in the case of WDM equipment.

⁵ The model should account for other contributions to service reliability, such as those from imperfect fault coverage, system impairing faults, and common cause failures [7].

6. When to use the 50% confidence data

There is a residual question concerning the point at which modellers should start to use the 50% confidence data. Initially, when no service history has accrued, the only information available to the modeller is the guaranteed FRU failure rate, which should be converted into an 'expected' failure rate by application of the 'failure rate adjustment' factor. Figure 5 shows that initially the 50% confidence estimates are likely to be high compared to the true value, and also to the expected value. As service history and the number of failures increases, 50% confidence reliability estimates should replace the expected value, irrespective of whether the estimate is higher or lower than the expected value, because the width of the distribution of the estimate will now be fairly small. The question can be answered by considering the variation in 50% confidence failure rate produced either by having one more (or one less) failure for the same time in service, or by changing the time in service (e.g. a card failure limit (CFL) change of ± 1) for the same number of failures (Fig 6). Clearly, most variation is expected when there is little service history, and least variation when there is a large amount of service history.

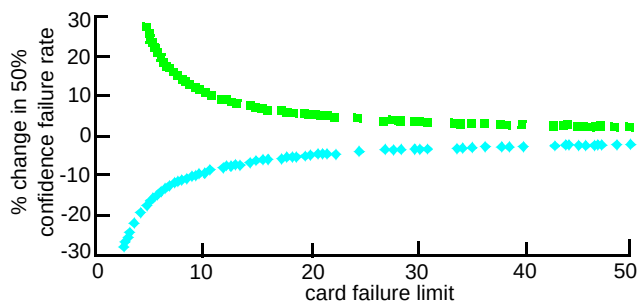


Fig 6 Percentage change in 50% confidence failure rate for a change of ± 1 in the CFL value as a function of CFL value.

The two methods actually produce very similar results, but the CFL approach was chosen, because any limit will be reached earlier than if the number of failures approach was chosen, since manufacturers generally beat their guarantees. Allowing a maximum 10% change in failure rate, from a CFL change of 1, gives a limit of CFL = 9. In other words, 50% confidence level data for FRU failure rates should be used when CFL ≥ 9.0 . One feature of this argument is that it is based entirely on mathematical concepts rather than on a peer comparison of FRUs, and should therefore be more readily acceptable to modellers and manufacturers.

7. Conclusions

In reviewing the concepts and techniques adopted by BT in order to better understand network reliability, this paper has highlighted four key points.

- On average, manufacturers exceed their FRU reliability guarantees by a factor of just over 2.
- In most circumstances, a confidence level of 50% should be used for FRU failure rates used in network reliability modelling. Table 2 indicates the possible consequences of not doing so.

Table 2 Possible consequences of failure to use 50% confidence level.

Confidence level in FRU failure rate	Resulting confidence level in failure rate of series in FRUs	Potential impact of network modelling result
<50%	<<50%	Underestimates network failure rate. Could result in high level of compensation claims
= 50%	= 50%	Gives average network failure rate. Should result in a design that achieves a balance between too little resilience and too much resilience.
>50%	>> 50%	Overestimates likely network failure rate. Could result in unnecessarily resilient designs.

- ISM schemes should be based on simple measures of FRU reliability. Additionally, for resilient networks, the scheme should include some measure of service reliability.
- It is recommended that consideration is given to the use of 50% confidence limits for infrastructure failure rates.

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David Ford graduated from Manchester University in 1977, obtaining a first class honours degree in Physics. After 8 years reliability experience in the automotive industry he joined BT in 1985, working on component life testing and failure analysis at the materials and component centre in Birmingham. He joined the Network Transport Engineering Centre in 1994 where he has worked on reliability aspects of various projects including HomeHighway and WDM. Recently, he has worked on core transport design for the CBP Midband programme. He was awarded an MSc with distinction in Quality Improvement and System Reliability by City University in 1995 and is a member of the IEE and a Chartered Engineer.